# Tangente Lösungen · mehr

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|---|---|---|
| **1.** | **$g(t) \approx -0{,}597t + 93{,}77$; ab 90 Tagen** | $m = h'(60) = -12e^{-3} \approx -0{,}597$; $h(60) \approx 57{,}92 \Rightarrow g(t) \approx -0{,}597t + 93{,}77$; $g(t) \leq 40 \Rightarrow t \geq 90 \Rightarrow$ ab 90 Tagen |
| **2.** | **$x \approx 4{,}44$ ($(2 - 8\sqrt{2})/3 < 2$ entfällt)** | $f'(x) = -\frac{3}{50}x^2 + \frac{2}{25}x + \frac{2}{5} = -\frac{32}{75} \Rightarrow 9x^2 - 12x - 124 = 0 \Rightarrow x = (2 \pm 8\sqrt{2})/3 \Rightarrow x = (2 + 8\sqrt{2})/3 \approx 4{,}44$ ($(2 - 8\sqrt{2})/3 < 2$ entfällt) |
| **3.** | **t: $y = -2x + 4$** | $f(0) = 4$; $f'(0) = -2 \Rightarrow$ t: $y = -2x + 4$ |
| **4.** | **t: $y = 2x + 1$** | $r(0) = 1 = s(0) \Rightarrow$ Punkt $(0\|1)$; $r'(x) = -2x + 2 \Rightarrow r'(0) = 2$; $s'(x) = 2e^{2x} \Rightarrow s'(0) = 2 \Rightarrow$ gleiche Steigung $\Rightarrow$ t: $y = 2x + 1$ |